

Ahana Kaur

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Professional Summary

Python developer and ML engineer specializing in computer vision, geospatial AI, and deployable inference systems. Built end-to-end pipelines spanning satellite imagery analysis, semantic segmentation, NLP, and offline AI applications – from prototype through deployment. Comfortable operating across the full stack: data pipelines, model training, REST APIs, and real-world constraints. Fast learner who ships.

Education

Bennett University

Master of Computer Applications (MCA) – Currently 4th Semester

Greater Noida, India

Aug 2024 - June 2026

- Relevant Coursework: Statistical Machine Learning, Deep Learning, NLP, Soft Computing, Image & Video Processing

Presidency University

Bachelor of Computer Applications (BCA)

Bangalore, India

October 2021 - May 2024

- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Programming in Python/C

Experience

CoGrad (in partnership with IIT Delhi – IHFC)

AI Trainer Intern

India, On-site

Mar 2026 - May 2026

- Delivered AI/ML training on the **KVS-IHFC STEM Teacher Training Programme** – a national upskilling initiative for Kendriya Vidyalaya teachers, in partnership with IIT Delhi's IHFC.
- Led end-to-end delivery of **4 training cohorts**, covering AI/ML concepts, prompt engineering, and deployment tools (NotebookLM, Gamma, Prezi, agentic IDEs); handled on-ground technical troubleshooting across all sessions.
- Coordinated participant groups, tracked attendance, and authored cohort completion reports across multiple venue deployments across India.

GJ Map Solutions Pvt. Ltd.

Python Developer Intern – Geospatial AI & Computer Vision

Remote

Feb 2026 - Apr 2026

- Analyzed **Sentinel-2 satellite imagery** for land-use/land-cover (LULC) classification; processed high-resolution aerial and drone imagery for geospatial mapping workflows.
- Built and trained CV models for **road detection, tree segmentation, and ship detection** in coastal imagery using YOLO (object detection) and U-Net (semantic segmentation).
- Developed ML classification pipelines for terrain recognition from aerial imagery; handled dataset preprocessing, annotation, and feature extraction end-to-end.
- Executed spatial analysis and visualization in **ArcGIS and QGIS**, including 360° imagery processing and mobile image geotagging workflows.
- Scripted geospatial ML tooling in Python (Rasterio, Shapely, PyProj, OpenCV, PyTorch) – from requirement analysis through prototype delivery.

Tata Group (via Forage)

Data Analytics Job Simulation

Remote

Dec 2025

- Applied GenAI-assisted EDA to assess data quality and surface customer risk indicators across a large dataset.
- Designed a no-code predictive delinquency risk framework and AI-driven collections strategy, incorporating ethical AI and regulatory compliance considerations.

Technical Skills

Core Languages: Python, SQL (MySQL, SQLite)

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers, NLP (NLTK, spaCy), Whisper, Pyannote, LLM Inference, Prompt Engineering

Computer Vision: OpenCV, CNNs, Object Detection (YOLO), Segmentation (U-Net), Real-Time Video Processing, MediaPipe, Image Augmentation

Geospatial & Remote Sensing: Rasterio, Shapely, PyProj, ArcGIS, QGIS, Sentinel-2 Processing, LULC Classification, OSM Road Extraction, Drone Image Analysis

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis

Web & APIs: Flask, FastAPI, REST API Development, Firebase, Kivy, Tkinter

MLOps & Tools: Git & GitHub, Docker, Jenkins, Linux, Google Colab, VS Code, Jupyter Notebook

Soft Skills: Systems thinking, structured problem-solving, self-directed learning, technical communication, adaptability

Selected Projects

NoiseFloor – Intelligent Job Discovery System | *Python, NLP, SQLite, Twilio, Flask, Render* | [Deployed]

- Architected and shipped a full-stack ML system reducing irrelevant job listings by **70–80%** using semantic resume-JD matching and legitimacy scoring.
- Built rule-based risk detection for job scams and suspicious postings – providing **explainable signals**, not binary filters.
- Deployed to Render with event-driven webhooks and scheduled scraping for **24/7 autonomous operation**; real-time WhatsApp alerts via Twilio with human-in-the-loop feedback capture.
- Persistent SQLite deduplication eliminated repeated daily alerts and prioritized newly posted roles.

Offline AI Meeting Transcription & Summarization System | *Whisper, Pyannote, Transformers, PyTorch* | [In Progress]

- Built multilingual ASR pipeline achieving **90%+ transcription accuracy**; reduced inference latency **35%** via chunking, silence removal, and denoising.
- Integrated Pyannote speaker diarization for **up to 6 speakers**; generated abstractive summaries condensing 30–60 min meetings into 5–7 key insights. Runs **fully offline**, zero cloud cost.

End-to-End MLOps Pipeline for ML API Deployment | *Jenkins, Docker, FastAPI, Scikit-learn*

- Designed CI/CD pipelines in Jenkins automating model testing, validation, and deployment; Dockerized inference APIs with **100% reproducible builds**.
- Deployed FastAPI prediction services handling **concurrent real-time requests** with structured JSON responses.

Real-Time Emotion Recognition System | *TensorFlow, OpenCV, CNNs*

- Trained CNN classifiers achieving **~70% validation accuracy** across 7 emotion classes; real-time webcam inference at **15–20 FPS** on consumer hardware.
- Logged emotion predictions at 5-second intervals for temporal behavior analysis and pattern tracking.

AI Tutor / Educational Assistant | *BART, Hugging Face, Flask, SQLite*

- Document ingestion pipelines (PDF/PPT/image) with BART-based abstractive summarization reducing content length by **60–70%**; auto-generated personalized quizzes.

Neural Machine Translation (English-Hindi) | *MarianMT, Hugging Face, PyTorch*

- Transformer encoder-decoder translation with fluent sentence-level output and interactive source/output visualization interface.

Certifications

- Generative AI: Introduction and Applications – IBM
- Programming for Everybody (Python) – University of Michigan
- Computer Networking – Google

Areas of Interest

- Computer Vision systems, Geospatial AI & Remote Sensing, Real-time ML inference, Python automation, Multimodal AI, Responsible AI & evaluation, Edge / offline deployment, AI for accessibility